Yeast Fermentation

Teacher guide | Flexible classroom investigation | Biology 2e, Chapter 7

1. Choose the equipment and organize the class

Question: How does the available sugar affect gas accumulation in a yeast culture? Use no added sugar, glucose, and sucrose. Students work in pairs or small teams, then pool a complete comparison set.

Option	Measurement	Use and limitation
Simple classroom lab	Balloon circumference (cm)	Matched flasks or bottles and balloons. Relative indicator of gas accumulation; not gas volume.
Quantitative lab	Collected gas volume (mL)	Suitable gas syringe or approved gas-collection apparatus. Check leaks, capacity and unobstructed movement.
Sensor lab	Pressure (kPa) OR CO_2 (ppm or %)	Use school-approved equipment and its protocol. Keep headspace and temperature fixed; do not treat pressure or concentration as volume.

One full comparison set = 9 independent flasks: N1-N3 (no added sugar), G1-G3 (glucose), S1-S3 (sucrose). Assign each pair one or two flask IDs. Each trial is a fresh mixture in a separate flask; repeated readings from one flask are not extra trials.

Students in a pooled group	Example allocation for all 9 flasks
18: nine pairs	Each pair prepares 1 flask.
12: six pairs	Three pairs prepare 2; three pairs prepare 1.
10: five pairs	Four pairs prepare 2; one pair prepares 1.
9: three pairs and one trio	Each team prepares 2; the teacher prepares the ninth flask. Alternatively assign the trio 3.

Roles: prepare/mix; measure/time; record/share. Pairs can combine roles and swap. Teams with two flasks must record separate start times and track elapsed time for each.

Demo option: distribute observation and analysis roles while the teacher prepares the apparatus. If only one flask per treatment is available, report n = 1; do not label repeated time readings as three trials. Pool additional matched, independent class setups if available.

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Teacher guide | Flexible classroom investigation | Biology 2e, Chapter 7

2. Stock solutions and flask sizes

Choose one flask size and recipe for an entire pooled set. These are starting volumes to pilot, not a guarantee of visible balloon inflation in one period. Use identical balloons that fit securely; small samples may make too little gas to inflate stiff balloons.

Flask capacity	Treatment stock*	Yeast suspension	Final liquid
50 mL	10 mL	10 mL	20 mL
100 mL	20 mL	20 mL	40 mL
Teacher-selected: ____ mL	____ mL	____ mL	____ mL

*Treatment stock is plain water for N, glucose solution for G, or sucrose solution for S. These examples leave 60% of flask capacity empty for headspace/foam; lower the fill if your pilot requires it. Do not fill a 50 mL flask with 50 mL of culture.

Optional stock recipe matching the virtual concentration

For each sugar stock: dissolve 80 g of glucose OR sucrose and bring to a final volume of 1,000 mL with water (8% mass/volume). For yeast: disperse 20 g dry baker's yeast and bring to 1,000 mL (2% mass/volume suspension). Keep a separate plain-water stock. Equal stock and yeast volumes give 4% sugar and 1% yeast in the final mixture. The no-sugar control receives no sugar.

This matches the virtual lab's mass concentrations, not its apparatus or volume. Use the same glucose product throughout; record if it is dextrose monohydrate. Equal sugar mass concentration is not equal molarity. A teacher may use another piloted recipe and record it below.

Prepare sugar stocks ahead in labeled clean containers. Prepare yeast close to use following its package hydration guidance; do not add sugar to the shared yeast stock. Gently resuspend immediately before every aliquot because yeast settles. Use separate labeled transfer tools and equilibrate all stocks to the same chosen test temperature.

For one 9-flask set	50 mL flasks	100 mL flasks
Plain water / glucose / sucrose (each)	30 mL	60 mL
Yeast suspension (all 9 flasks)	90 mL	180 mL
Complete sets from 1 L of EACH stock*	11 sets	5 sets

*Before transfer losses; yeast is limiting. For B sets and V mL of each stock per flask: each treatment stock = 3 x B x V; yeast = 9 x B x V. Allow extra for spills and the pilot. Optional controls need additional stock.

Chosen sugar stock: ____ g / ____ mL final stock. Yeast: ____ g / ____ mL. Chosen test temperature: ____ C. Final sugar concentration: ____ %.

Yeast Fermentation

3. Run, measure, and interpret

Classroom sequence

Allow about 10-15 minutes to plan and prepare, 30 minutes to observe, and 15-20 minutes to pool and interpret. Adjust after a pilot; use the next lesson for analysis if needed. A shared 30 C water bath is one possible test condition; keep all treatments at the same measured temperature.

1. Assign nine flask IDs using page 4. Students label their flasks, record their prediction, and copy the teacher-approved volumes, temperature, measurement and schedule.

2. Dispense treatment stock first. Gently resuspend the common yeast stock, add the assigned volume last, mix each flask consistently, fit the balloon promptly, and start that flask's timer. Record initial observations. Do not inflate balloons by mouth; use the same teacher-selected preparation for every balloon.

3. Record at 0, 5, 10, 15, 20, 25 and 30 elapsed minutes, or a common teacher-selected schedule. With balloons, measure the widest circumference gently with string/ruler or flexible tape. Do not squeeze. Use the same location/method each time.

4. For a limp balloon, record NI (not inflated), not a fabricated zero. Record M for missing and L for a suspected leak, with notes. A visibly inflated balloon may be measured even if small. Report time to first visible inflation separately if useful.

5. Share raw readings and flask IDs. Average only comparable numeric readings and report n at each time. If some trials are NI, report how many are uninflated and any numeric-only mean separately; do not imply it represents all three flasks. Do not silently discard unusual results.

Interpretation and alternative equipment

Circumference is a proxy affected by balloon elasticity, leaks, temperature and gas dissolved in the liquid. Do not convert it to mL using a spherical-balloon formula. A cm/min change is a circumference-change rate, not a gas-production rate. For gas syringes use mL and baseline-correct readings; for sensors use the manufacturer's calibration and operating procedure.

Compare only matching apparatus, recipes, temperatures and units in a pooled mean. Keep different flask sizes, measurement methods and virtual results separate. Gas accumulation supports a claim about yeast activity; it does not directly measure ATP yield or establish oxygen absence. Do not require glucose to beat sucrose.

Practical checks

Use eye protection and school lab procedures. Check glassware, balloon fit and latex allergies. Never cap a gas-producing culture in a rigid closed container; never obstruct a gas syringe. Pressure sensors require a rated system and its specified pressure limit. Do not taste cultures. Teacher handles warm water, stops foaming/overfilled setups and directs cleanup.

If nothing inflates: check yeast freshness, temperature, settling, leaks and balloon stiffness; allow more time if practical. A sugar-without-yeast control can help assess physical gas expansion. Record it separately from the nine required flasks.

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4. Assignment roster and lesson settings

Class/date: _____ Pooled set ID: _____ Students: ____

Method/unit: _____ Flask capacity: ____ mL Balloon type: _____

Treatment stock per flask: ____ mL Yeast suspension per flask: ____ mL

Temperature: ____ C Reading interval: ____ min Final time: ____ min

Flask ID	Treatment	Pair/team or demo	Start time
N1	No added sugar		
N2	No added sugar		
N3	No added sugar		
G1	Glucose		
G2	Glucose		
G3	Glucose		
S1	Sucrose		
S2	Sucrose		
S3	Sucrose		

If fewer than three valid independent trials remain, report the actual sample size and reason. Add a fresh replacement if time allows; retain the original record with its flag.

Print and share

Print worksheet pages 1-2 for each pair/team and pages 3-4 for each student or pooled group. Page 2 records up to two flasks; copy it for additional flasks. Page 3 combines all nine. For other collection methods, change the measurement/unit fields and provide the equipment protocol.

The website's virtual lab uses gas volume (mL). Keep balloon and sensor results on this worksheet or a separate class spreadsheet; do not enter cm, kPa or ppm into its mL fields. No login is needed to download these handouts.

Method and reading connections

Science World: Yeast-Inflated Balloons https://www.scienceworld.ca/resource/yeast-inflated-balloons/ Science Buddies: Yeasty Beasties https://www.sciencebuddies.org/science-fair-projects/project-ideas/MicroBio_p011/microbiology/yeast-activity-balloons OpenStax Biology 2e, 7.5: Metabolism without Oxygen https://openstax.org/books/biology-2e/pages/7-5-metabolism-without-oxygen

This guide adapts the measurement approach to pooled classroom trials. Small-flask recipes and volumes above are proposed pilot settings, not validated performance claims.